

Exp 2 - Computer Networks

Type/Category	Advantages	Disadvantages	Applications
Twinaxial Cable	High data transmission capability; good resistance to interference; lower latency vs. some optical connections; suitable for short-distance high-speed connections	Limited transmission distance; heavier than fibre-optic cables; less suitable for long-distance communication	Data centres; server connections; high-speed switch connections
Direct Attach Copper (DAC) Cable	Low cost vs. many optical solutions; low latency; simple to install; suitable for short-distance connections	Limited distance; heavier than optical alternatives; can consume more power in some applications	Server-to-switch connections; data centres; high-speed Ethernet connections
Active Optical Cable (AOC)	High bandwidth; lightweight; resistant to electromagnetic interference; suitable for high-speed connections	More expensive than ordinary copper cables; generally less flexible in terms of repair; requires compatible optical interfaces	Data centres; high-performance computing; high-speed switch and server connections
Ribbon Fibre Cable	High fibre density; allows many fibres in a small space; efficient for large-scale installations	More complex to handle than ordinary cables; requires specialised installation equipment; damage can affect multiple fibres	Data centres; telecommunications; high-density fibre networks
Armoured Fibre Optic Cable	Strong mechanical protection; better resistance to crushing and impact; suitable for harsh environments	Heavier than standard fibre cable; more expensive; installation can be more difficult	Outdoor networks; industrial environments; areas with risk of physical damage

Plastic Optical Fibre (POF)	Lightweight; flexible; easy to handle; less complicated to install than some glass-fibre systems	Lower performance over long distances; higher transmission losses than glass fibre; not normally used for long-distance backbone networks	Short-distance networking; industrial systems; consumer and multimedia applications
Single-Pair Ethernet (SPE) Cable	Requires less cabling; smaller and lighter; useful for sensors/industrial devices; suitable for space-constrained installations	Not intended to replace every conventional Ethernet connection; distance/speed depend on the specific SPE standard; requires compatible equipment	Industrial automation; IoT devices; sensors; building automation
Telephone / DSL Cable	Uses existing telephone infrastructure; relatively inexpensive; easy to deploy where telephone wiring already exists	Lower bandwidth than modern Ethernet/fibre; performance decreases with distance; susceptible to interference	Telephone networks; DSL internet connections; older residential networking systems